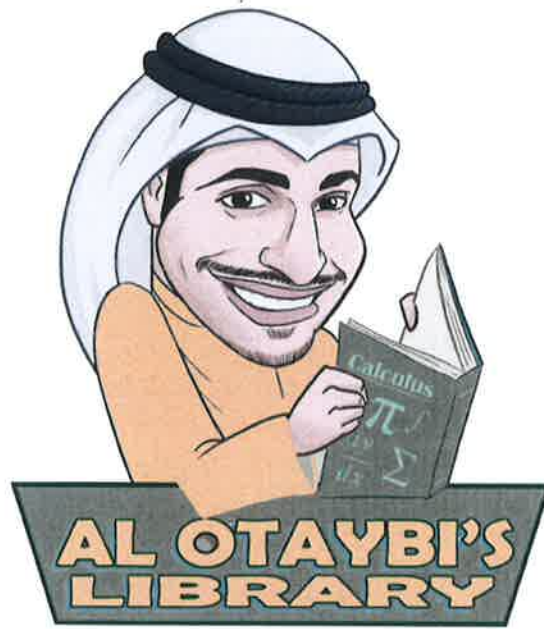


Kuwait University

2021/2022

Spring course



Calculus B

Second

مراجعة

اختبارات سابقة

(مراجعة السكند)

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Math 102: Calculus II

Second Exam • 17 December 2016 • Duration: 90 Minutes

In questions 1-5, select the correct answer. No justification is required (5 points each).

1. The sequence $\left\{ \frac{1}{\sqrt[5]{n^2 + 1}} \right\}$ is _____.

- (a) increasing and divergent (b) increasing and convergent
(c) decreasing and divergent (d) decreasing and convergent
(e) alternating and divergent

2. Which of the following is true for a series $\sum_{n=1}^{\infty} a_n$?

- (a) If the series is convergent, then it is also absolutely convergent.
(b) If the series is alternating, then it is convergent.
(c) If $\lim_{n \rightarrow \infty} a_n = L$, then $\sum_{n=1}^{\infty} a_n = L$.
(d) If $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.
(e) If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.

3. The sum of the series $\sum_{n=1}^{\infty} 2^{4-3n}$ is _____.

- (a) 16 (b) $\frac{128}{7}$ (c) $\frac{16}{7}$ (d) 0 (e) The series diverges

4. The n th partial sum of the series $\sum_{n=1}^{\infty} \frac{2}{(2n-3)(2n-1)}$ is _____.

- (a) $-1 + \frac{2}{(2n-3)(2n-1)}$ (b) $-1 - \frac{1}{2n-1}$ (c) $1 - \frac{1}{2n-3}$
(d) $1 + \frac{1}{3} - \frac{1}{2n-3} - \frac{1}{2n-1}$ (e) $-1 + 1 - \frac{1}{2n-3} - \frac{1}{2n-1}$

5. The binomial coefficient $\binom{5/2}{3}$ equals _____.

- (a) 125/8 (b) 5/6 (c) -5/9 (d) 5/16 (e) $\sqrt{3^5}$

6. [10+5 pts] Consider the sequence $a_n = \cos\left(\frac{2n\pi}{n^2 + n + 2}\right)$, $n \geq 1$.

(a) Is $\{a_n\}$ convergent or divergent?

(b) What can be said about convergence of the series $\sum_{n=1}^{\infty} a_n$?

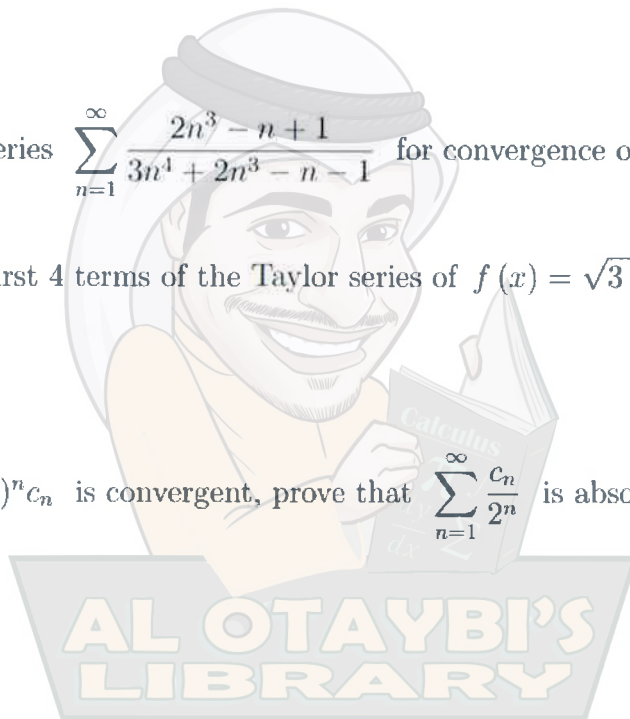
7. [15 pts] Find the radius and interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-1)^n (x+3)^n}{(2n)!}$.

8. [15 pts] Determine whether the series $\sum_{n=1}^{\infty} (-1)^n \tan\left(\frac{1}{n}\right)$ is absolutely convergent, conditionally convergent, or divergent.

9. [10 pts] Test the series $\sum_{n=1}^{\infty} \frac{2n^3 - n + 1}{3n^4 + 2n^3 - n - 1}$ for convergence or divergence.

10. [10 pts] Find the first 4 terms of the Taylor series of $f(x) = \sqrt{3+x^2}$ at $x = 1$.

11. [10 pts] If $\sum_{n=1}^{\infty} (-1)^n c_n$ is convergent, prove that $\sum_{n=1}^{\infty} \frac{c_n}{2^n}$ is absolutely convergent.





Multiple choice



The sequence $\left\{ \frac{1}{\sqrt[5]{n^2 + 1}} \right\}$ is :

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- a) increasing and divergent. b) increasing and convergent.
c) decreasing and divergent. d) decreasing and convergent.
e) alternating and divergent.

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[5]{n^2 + 1}} = \frac{1}{\infty} = 0$$

$\Rightarrow \therefore a_n$ is convergent.

$$f(x) = \frac{1}{\sqrt[5]{x^2 + 1}} = (x^2 + 1)^{-\frac{1}{5}}$$

$$f'(x) = \frac{-1}{5} (x^2 + 1)^{-\frac{6}{5}} 2x$$

$$= \frac{-2x}{5 (x^2 + 1)^{\frac{6}{5}}} < 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore a_n$ is decreasing

(d)
✓



Multiple choice



Which of the following is true for the series $\sum_{n=1}^{\infty} a_n$?

(a) If the series is convergent, then it also absolutely convergent.

(b) If the series is alternating, then it is convergent.

(c) If $\lim_{n \rightarrow \infty} a_n = L$, then $\sum_{n=1}^{\infty} a_n = L$.

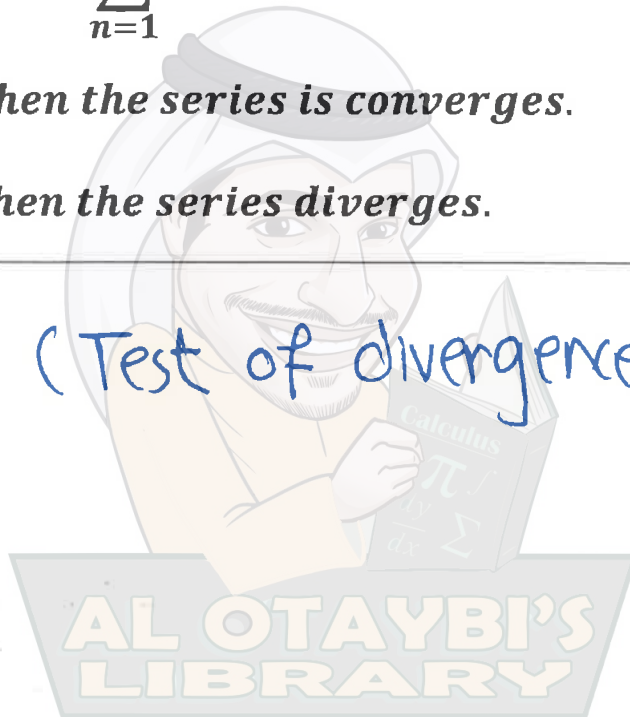
(d) If $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.

(e) If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.

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(e) ✓

(Test of divergence)





Multiple choice



The sum of the series $\sum_{n=1}^{\infty} 2^{4-3n}$ is

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- a) 16. b) $\frac{128}{7}$ c) $\frac{16}{7}$ d) 0 e) The series diverges.

$$n=1 \rightarrow a_1 = 2^{4-3} = 2$$

$$n=2 \rightarrow a_2 = 2^{4-6} = 2^{-2} = \frac{1}{2^2}$$

$$\therefore r = \frac{a_2}{a_1} = \frac{\frac{1}{2^2}}{2} = \frac{\frac{1}{4}}{2} = \frac{1}{8}$$

$|r| < 1 \rightarrow \therefore \sum_{n=1}^{\infty} 2^{4-3n}$ is convergent
(Geometric Series)

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{2}{1-\frac{1}{8}} = \frac{2}{\frac{8}{8}-\frac{1}{8}} = \frac{2}{\frac{7}{8}} \\ &= \frac{16}{7} \end{aligned}$$

(c)



Multiple choice



The n th partial sum of the series $\sum_{n=1}^{\infty} \frac{2}{(2n-3)(2n-1)}$ is

a) $-1 + \frac{2}{(2n-3)(2n-1)}$

b) $-1 - \frac{1}{2n-1}$

c) $1 - \frac{1}{2n-3}$

d) $1 + \frac{1}{3} - \frac{1}{2n-3} - \frac{1}{2n-1}$

e) $-1 + 1 - \frac{1}{2n-3} - \frac{1}{2n-1}$

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Let $f(x) = \frac{2}{(2x-3)(2x-1)}$

$$\frac{2}{(2x-3)(2x-1)} = \frac{A}{2x-3} + \frac{B}{2x-1}$$

* $(2x-3)(2x-1)$

$$2 = A(2x-1) + B(2x-3)$$

$x = \frac{1}{2} \rightarrow 2 = 0 + B(2 \cdot \frac{1}{2} - 3) \rightarrow 2 = -2B \rightarrow B = -1$

$x = \frac{3}{2} \rightarrow 2 = A(2 \cdot \frac{3}{2} - 1) + 0 \rightarrow 2 = 2A \rightarrow A = 1$

$$\therefore a_n = \frac{1}{2n-3} - \frac{1}{2n-1}$$

$$S_n = a_1 + a_2 + \dots + a_{n-1} + a_n$$

$$= \left[\frac{1}{-1} - \frac{1}{1} \right] + \left[\frac{1}{1} - \frac{1}{3} \right] + \dots + \left[\frac{1}{2(n-1)-3} - \frac{1}{2(n-1)-1} \right]$$

$$\therefore S_n = -1 - \frac{1}{2n-1} \quad (b)$$



Multiple choice



The binomial coefficient $\binom{5/2}{3}$ equals

Dec 2016

- a) $125/8$ b) $5/6$ c) $-5/9$ d) $5/16$ e) $\sqrt{3^5}$

$$\text{binomial coeff.} = \frac{\left(\frac{5}{2}\right)\left(\frac{5}{2}-1\right)\left(\frac{5}{2}-2\right)}{3!}$$

$$= \frac{\frac{5}{2} \left(\frac{3}{2}\right) \left(\frac{1}{2}\right)}{3!}$$

$$= \frac{15}{8} \cdot \frac{1}{3(2)} = \frac{15^5}{8(3)(2)} = \frac{5}{16} \quad \text{(d)}$$

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Consider the sequence $a_n = \cos\left(\frac{2n\pi}{n^2+n+2}\right)$, $n \geq 1$.

(a) Is $\{a_n\}$ convergent or divergent.

(b) What can be said about convergence of the series

$$\sum_{n=1}^{\infty} a_n ?$$

$$(a) \lim_{n \rightarrow \infty} \cos\left(\frac{2n\pi}{n^2+n+2}\right)$$

$$\lim_{n \rightarrow \infty} \cos\left(\frac{2\cancel{n}\pi}{\cancel{n}^2\left[1+\frac{1}{n}+\frac{2}{n^2}\right]}\right)$$

$$\lim_{n \rightarrow \infty} \cos\left(\frac{2\pi}{n\left[1+\frac{1}{n}+\frac{2}{n^2}\right]}\right) = \cos\left(\frac{2\pi}{\infty[1+0+0]}\right) = \cos(0) = 1$$

$\therefore a_n$ is convergent.

$$(b) \sum_{n=1}^{\infty} a_n$$

$$\sum_{n=1}^{\infty} \cos\left(\frac{2n\pi}{n^2+n+2}\right)$$

$$\lim_{n \rightarrow \infty} \cos\left(\frac{2n\pi}{n^2+n+2}\right) = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Test of divergence)

Find the radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x+3)^n}{(2n)!}$$

$$U_n = \frac{(-1)^n (x+3)^n}{(2n)!} \longrightarrow U_{n+1} = \frac{(-1)^{n+1} (x+3)^{n+1}}{(2(n+1))!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (x+3)^{n+1}}{(2n+2)!} * \frac{(2n)!}{(-1)^n (x+3)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(x+3)^{\cancel{n}} (x+3)}{(2n+2)(2n+1)\cancel{(2n)!}} * \frac{\cancel{(2n)!}}{(x+3)^{\cancel{n}}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(x+3)}{(2n+2)(2n+1)} \right| = \frac{x+3}{\infty} = 0$$

∴ The radius of convergence $\longrightarrow R = \infty$

The interval of convergence $\longrightarrow I = (-\infty, \infty)$

Determine whether the series

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$$\sum_{n=1}^{\infty} (-1)^n \tan\left(\frac{1}{n}\right) \rightarrow a_n$$

is absolutely convergent, conditionally convergent or divergent.

$$|a_n| = |(-1)^n \tan\left(\frac{1}{n}\right)| = \left|\tan\left(\frac{1}{n}\right)\right|$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$$

$$\text{Let } b_n = \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ divergent (Harmonic)}$$

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n}$$

$$\lim_{n \rightarrow \infty} \tan\left(\frac{1}{n}\right) \neq n$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1 > 0$$

$$\therefore \sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right) \text{ divergent (L.C.T)}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \tan\left(\frac{1}{n}\right) = \tan 0 = 0$$

$$\text{Let } f(x) = \tan\left(\frac{1}{x}\right)$$

$$f'(x) = \sec^2\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right) < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing on $[1, \infty)$

$\therefore \sum_{n=1}^{\infty} a_n$ is conditionally convergent

Test the series

$$\sum_{n=1}^{\infty} \frac{2n^3 - n + 1}{3n^4 + 2n^3 - n - 1} \rightarrow a_n$$

for convergence or divergence.

Let $b_n = \frac{n^3}{n^4} = \frac{1}{n}$

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent (Harmonic)

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$

$\lim_{n \rightarrow \infty} \frac{2n^3 - n + 1}{3n^4 + 2n^3 - n - 1} * n$

$\lim_{n \rightarrow \infty} \frac{n^3 \left[2 - \frac{1}{n^2} + \frac{1}{n^3} \right]}{n^4 \left[3 + \frac{2}{n} - \frac{1}{n^3} - \frac{1}{n^4} \right]} * n$

$= \frac{2 - 0 + 0}{3 + 0 - 0 - 0} = \frac{2}{3} > 0$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (L.C.T)

Find the first 4 terms of the Taylor series of

$$f(x) = \sqrt{3+x^2} \text{ at } x = 1$$

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$$f(x) = \sqrt{3+x^2} \longrightarrow f(1) = \sqrt{4} = \boxed{2}$$

$$f'(x) = \frac{2x}{2\sqrt{3+x^2}} \longrightarrow f'(1) = \frac{1}{\sqrt{4}} = \boxed{\frac{1}{2}}$$

$$f''(x) = \frac{\sqrt{3+x^2}(1) - x \cdot \frac{2x}{2\sqrt{3+x^2}}}{3+x^2} = \frac{\frac{3+x^2-x^2}{\sqrt{3+x^2}}}{3+x^2} \longrightarrow f''(1) = \frac{3}{(4)^{\frac{3}{2}}} = \frac{3}{2^3} = \boxed{\frac{3}{8}}$$

$$f'''(x) = \frac{-3 \cdot \frac{3}{2} (3+x^2)^{-\frac{1}{2}} \cdot 2x}{(3+x^2)^3} \longrightarrow f'''(1) = \frac{-9(4)^{-\frac{1}{2}}(1)}{(4)^3} = \frac{-9(2)}{64} = \boxed{\frac{-9}{32}}$$

$$T_3(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3$$

$$= 2 + \frac{\frac{1}{2}}{1}(x-1) + \frac{\frac{3}{8}}{2!}(x-1)^2 + \frac{\frac{-9}{32}}{3!}(x-1)^3$$

if $\sum_{n=1}^{\infty} (-1)^n C_n$ is absolutely convergent.

Prove $\sum_{n=1}^{\infty} \frac{C_n}{2^n}$ is absolutely convergent.

Sol #1

$\therefore \sum_{n=1}^{\infty} (-1)^n C_n$ is absolutely convergent

$\therefore \sum_{n=1}^{\infty} |(-1)^n C_n|$ is convergent $\rightarrow \therefore \sum_{n=1}^{\infty} |C_n|$ is convergent

$\Rightarrow \therefore$ The interval of convergence contain $(-1, 1)$

$\sum_{n=1}^{\infty} \frac{C_n}{2^n}$ abs. conv. ?

$\sum_{n=1}^{\infty} \left| \frac{C_n}{2^n} \right| \rightarrow \sum_{n=1}^{\infty} |C_n| \left(\frac{1}{2} \right)^n$

$\therefore \frac{1}{2} \in (-1, 1) \rightarrow \therefore \sum_{n=1}^{\infty} |C_n| \left(\frac{1}{2} \right)^n$ is conv.

$\therefore \sum_{n=1}^{\infty} \frac{C_n}{2^n}$ is absolutely convergent.

Sol #2

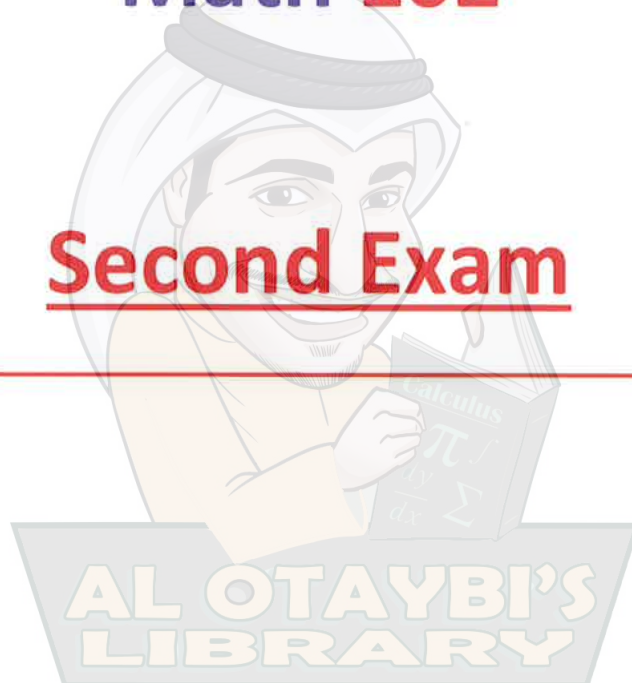
B.C.T..

Calculus B

Fall Semester 2018

Math 102

Second Exam





Multiple choices



1) The general term of the sequence

$$\left\{-1, \frac{2}{9}, -\frac{4}{25}, \frac{8}{49}, \dots\right\}, \quad n = 0, 1, 2, \dots$$

a) $a_n = \frac{(-1)^n 2^n}{(2n+1)^2}$ b) $a_n = \frac{(-1)^{n+1} 2^n}{(2n+1)^2}$ c) $a_n = \frac{(-1)^n 2^n}{(2n)^2}$
 d) $a_n = \frac{(-1)^{n+1} 2^n}{(2n)^2}$ e) $a_n = \frac{2^n}{(2n+1)^2}$

Fall 2018

2) If a convergent sequence defined by

$$a_1 = 1 \text{ and } a_{n+1} = \frac{1}{2 + a_n} \quad n \in \mathbb{N} \text{ then}$$

$$\lim_{n \rightarrow \infty} a_n =$$

- A) 0 B) -1 C) $-1 + \sqrt{2}$
 D) $-\sqrt{2} - 1$ E) None of these

3) The sum of the series of

$$\sum_{n=0}^{\infty} \left(\frac{-2}{3}\right)^{n+1} \text{ is}$$

- A) $\frac{-2}{3}$ B) $\frac{-2}{5}$ C) $\frac{-6}{5}$
 D) $\frac{3}{5}$ E) None of these

$$4) \sum_{n=4}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right) =$$

- A) $\frac{1}{2}$ B) 1 C) 0 D) $\sqrt{2}$ E) None of these

$$5) \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^3}\sqrt{n}} =$$

- A) is convergent B) has sum -1
 C) is divergent D) has sum 0
 E) None of these

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2) List the first three terms of the sequence

$$a_n = \frac{3\sqrt{n}}{\sqrt{n+2}}, \quad n = 1, 2, \dots$$

Determine whether the sequence is convergent or divergent, and find its limit if possible.

3) Use ratio test to determine if the series

$$\sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}$$

is convergent or divergent.

4) Determine if the series

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} \cos(n)}{(2n+5)^3}$$

a_n

Is absolute convergent, conditionally convergent or divergent.

5) Determine whether the series

$$\sum_{n=2}^{\infty} \frac{4}{3n(\ln n)^3}$$

a_n

is convergent or divergent.

6) Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

a) Show that the series is convergent.

b) How many terms of the series do we need to add in order to find the sum to the accuracy where $|\text{error}| < 0.001$? Justify your answer

7) Determine whether the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{9^n} \left(1 + \frac{2}{n}\right)^{n^2}$$

a_n

Is absolute convergent, conditionally convergent or divergent.



Multiple choices



1) The general term of the sequence

$$\left\{ -1, \frac{2}{9}, -\frac{4}{25}, \frac{8}{49}, \dots \right\}, \quad n = 0, 1, 2, \dots$$

a) $a_n = \frac{(-1)^n 2^n}{(2n+1)^2}$

b) $a_n = \frac{(-1)^{n+1} 2^n}{(2n+1)^2}$

c) $a_n = \frac{(-1)^n 2^n}{(2n)^2}$

d) $a_n = \frac{(-1)^{n+1} 2^n}{(2n)^2}$

e) $a_n = \frac{2^n}{(2n+1)^2}$

Fall 2018

(b) ✓



2) If a convergent sequence defined by

$$a_1 = 1 \text{ and } a_{n+1} = \frac{1}{2 + a_n} \quad n \in \mathbb{N} \text{ then}$$

$$\lim_{n \rightarrow \infty} a_n =$$

A) 0

B) -1

C) $-1 + \sqrt{2}$

D) $-\sqrt{2} - 1$

E) None of these

$\therefore a_n$ is convergent $\Rightarrow \lim_{n \rightarrow \infty} a_n = L$
 $\Rightarrow \lim_{n \rightarrow \infty} a_{n+1} = L$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2 + a_n}$$

$$L = \frac{1}{2 + L}$$

$$L(2 + L) = 1$$

$$2L + L^2 = 1$$

$$L^2 + 2L - 1 = 0$$

$$a = 1 \quad b = 2 \quad c = -1$$

$$\begin{aligned} L &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2(1)} \\ &= \frac{-2 \pm \sqrt{8}}{2} \quad \begin{matrix} \sqrt{4(2)} \\ \sqrt{4}\sqrt{2} \\ 2\sqrt{2} \end{matrix} \\ &= \frac{-2 \pm 2\sqrt{2}}{2} \\ &= \cancel{2} \frac{[-1 \pm \sqrt{2}]}{\cancel{2}} \end{aligned}$$

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$$\Rightarrow L = -1 + \sqrt{2}$$

$$\begin{aligned} n=1 &\rightarrow a_2 = \frac{1}{2+a_1} \rightarrow a_2 = \frac{1}{3} \\ n=2 &\rightarrow a_3 = \frac{1}{2+a_2} \rightarrow a_3 = \frac{2}{7} \end{aligned}$$

✓ (C)

3) The sum of the series of

$$\sum_{n=0}^{\infty} \left(\frac{-2}{3}\right)^{n+1} \text{ is}$$

A) $\frac{-2}{3}$

B) $\frac{-2}{5}$

C) $\frac{-6}{5}$

D) $\frac{3}{5}$

E) None of these

$$\sum_{n=0}^{\infty} \left(\frac{-2}{3}\right)^{n+1}$$

$$n=0 \longrightarrow a_0 = \left(\frac{-2}{3}\right)^{0+1} = \frac{-2}{3}$$

$$n=1 \longrightarrow a_1 = \left(\frac{-2}{3}\right)^2$$

$$r = \frac{a_1}{a_0} = \frac{\left(\frac{-2}{3}\right)^2}{\frac{-2}{3}} = \frac{-2}{3}$$

$$S = \frac{a}{1-r} = \frac{\frac{-2}{3}}{1 - \left(\frac{-2}{3}\right)} = \frac{\frac{-2}{3}}{\frac{3+2}{3}} = \frac{-2}{5}$$

B

$$4) \sum_{n=4}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right) =$$

A) $\frac{1}{2}$ B) 1 C) 0 D) $\sqrt{2}$ E) None of these

$$a_n = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$$

$$S_n = a_4 + a_5 + \dots + a_{n-1} + a_n$$

$$= \left(\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{5}} \right) + \left(\frac{1}{\sqrt{5}} - \frac{1}{\sqrt{6}} \right) + \dots + \left(\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n}} \right) + \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right)$$

$$\rightarrow S_n = \frac{1}{\sqrt{4}} - \frac{1}{\sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{1}{\sqrt{n+1}} \right)$$

$$= \frac{1}{2} - \frac{1}{\infty} = \frac{1}{2} - 0 = \frac{1}{2} \quad (A) \quad \#$$

Telescoping

$$5) \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}\sqrt[3]{n}} =$$

- A) is convergent B) has sum - 1
 C) is divergent D) has sum 0
 E) None of these

$\sum_{n=1}^{\infty} \frac{1}{n n^{\frac{1}{2}} n^{\frac{1}{3}}}$

$\sum_{n=1}^{\infty} \frac{1}{n^{1+\frac{1}{2}+\frac{1}{3}}}$

بالعزب أجمع الأسس
 $n^{1+\frac{1}{2}+\frac{1}{3}}$

$P = 1 + \frac{1}{2} + \frac{1}{3} > 1$

Convergent (P-series)

(A) ✓

2) List the first three terms of the sequence

$$a_n = \frac{3\sqrt{n}}{\sqrt{n} + 2}, \quad n = 1, 2, \dots$$

Determine whether the sequence is convergent or divergent, and find its limit if possible.

$$n=1 \longrightarrow a_1 = \frac{3\sqrt{1}}{1+2} = \frac{3}{3} = 1$$

$$n=2 \longrightarrow a_2 = \frac{3\sqrt{2}}{\sqrt{2}+2}$$

$$n=3 \longrightarrow a_3 = \frac{3\sqrt{3}}{\sqrt{3}+2}$$

$$\lim_{n \rightarrow \infty} \frac{3\sqrt{n}}{\sqrt{n} \left[1 + \frac{2}{\sqrt{n}}\right]} = \frac{3}{1+0} = 3$$

$\Rightarrow \therefore a_n$ is convergent to 3

3) Use ratio test to determine if the series

$$\sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}$$

is convergent or divergent.

$$a_n = \frac{10^n}{(n+1)4^{2n+1}} \rightarrow a_{n+1} = \frac{10^{n+1}}{(n+2)4^{2(n+1)+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{10^{n+1}}{(n+2)4^{2n+2+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{10^{n+1}}{(n+2)4^{2n+3}} \cdot \frac{(n+1)4^{2n+1}}{10^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{10 \cdot 10^1}{(n+2)4^{2n} \cdot 4^3} \cdot \frac{(n+1)4^{2n} \cdot 4^1}{10^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{10 \cancel{4} (n+1)}{4^3 \cdot 2 (n+2)} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{10 \cancel{4} (1 + \frac{1}{n})}{4^2 \cdot 2 (1 + \frac{2}{n})} \right| = \left| \frac{10 (1+0)}{16 (1+0)} \right| = \frac{5}{8} < 1$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent (Ratio)

4) Determine if the series

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} \cos(n)}{(2n+5)^3}$$

a_n

Is absolute convergent . conditionally convergent or divergent.

$$a_n = \frac{(-1)^{n+1} \cos(n)}{(2n+5)^3} \rightarrow |a_n| = \left| \frac{(-1)^{n+1} \cos(n)}{(2n+5)^3} \right|$$

$$= \frac{|\cos n|}{(2n+5)^3}$$

$$\sum_{n=0}^{\infty} |a_n|$$

$|a_n|$

$$\frac{|\cos n|}{(2n+5)^3}$$

$$0 < |\cos n| < 1$$

$$\div (2n+5)^3$$

$$0 < \frac{|\cos n|}{(2n+5)^3} < \frac{1}{(2n+5)^3}$$

$$< \frac{1}{(2n)^3}$$

$\Rightarrow \therefore \sum_{n=0}^{\infty} |a_n|$ is convergent (B.C.T)

$\Rightarrow \therefore \sum_{n=0}^{\infty} a_n$ is absolutely convergent

$$\sum \frac{1}{8n^3}$$

$P=3 > 1$
Convergent
(P-Series)

5) Determine whether the series

$$\sum_{n=2}^{\infty} \frac{4}{3n(\ln n)^3}$$

a_n

is convergent or divergent.

$$f(x) = \frac{4}{3x(\ln x)^3}$$

~~C.T.H.P.I~~
N.A.B.L
R.P.

1) $f(x)$ is continuous and positive on $[2, \infty)$

$$\begin{aligned} 2) f'(x) &= \frac{[3x(\ln x)^3] \cdot 0 - 4[3(\ln x)^3 + 3x \cdot 3(\ln x)^2 \cdot \frac{1}{x}]}{[3x(\ln x)^3]^2} \\ &= \frac{-4[3(\ln x)^3 + 9(\ln x)^2]}{[3x(\ln x)^3]^2} \\ &= \frac{-4 \cancel{3}(\ln x)^2[\ln x + 3]}{3^2 x^2 (\ln x)^6} = \frac{-4[\ln x + 3]}{3x^2(\ln x)^4} \end{aligned}$$

$\Rightarrow \therefore f(x)$ is decreasing $\forall x \in [2, \infty)$

$$\int_2^{\infty} \frac{4}{3x(\ln x)^3} dx$$

$$\begin{aligned} I_1 &= \int \frac{4}{3x(\ln x)^3} dx \quad \text{Let } u = \ln x \\ &= \frac{4}{3} \int \frac{1}{u^3} du = \frac{4}{3} \int u^{-3} du \\ &= \frac{4}{3} \int u^{-3} du = \frac{4}{3} \frac{u^{-2}}{-2} + C = -\frac{2}{3} \frac{1}{u^2} + C \\ &= -\frac{2}{3} \frac{1}{(\ln x)^2} + C \end{aligned}$$

$$\lim_{t \rightarrow \infty} \int_2^t \frac{4}{3x (\ln x)^3} dx$$

$$= \lim_{t \rightarrow \infty} \left[-\frac{2}{3} \frac{1}{(\ln x)^2} \right]_2^t$$

$$= -\frac{2}{3} \lim_{t \rightarrow \infty} \left[\frac{1}{(\ln t)^2} - \frac{1}{(\ln 2)^2} \right]$$

$$= -\frac{2}{3} \left[\frac{1}{\infty} - \frac{1}{(\ln 2)^2} \right]$$

$$= \frac{2}{3} \frac{1}{(\ln 2)^2}$$

$\Rightarrow \therefore \sum a_n$ is Convergent
(Integral test)

6) Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{4^n}$

a) Show that the series is convergent.

b) How many terms of the series do we need to add in order to find the sum to the accuracy where $|error| < 0.001$? Justify your answer

$$\sum_{n=1}^{\infty} \frac{(-1)^n \left(\frac{1}{4}\right)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n \frac{1}{4^n}}{n}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{4^n n}$$

a_n

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{4^n n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{4^n n} \right| = \frac{1}{\infty} = 0$$

Let $f(x) = \frac{1}{4^x x}$

$$f'(x) = \frac{0 - 4^x \ln 4 x + 4^x (1)}{[4^x x]^2}$$

$$= \frac{-4^x [\ln 4 x - 1]}{(4^x)^2 x} = -\frac{[\ln 4 x - 1]}{4^x x}$$

$\therefore f(x)$ is decreasing

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is convergent (A.S.T) $< 0 \quad \forall x \in [1, \infty)$

(b)

$$|a_{n+1}| < \text{error}$$

$$\frac{1}{4^{n+1}(n+1)} < 0.001$$

$$\frac{1}{4^{n+1}(n+1)} < \frac{1}{1000}$$

$$4^{n+1}(n+1) > 1000$$

$$4^n \cdot 4(n+1) > 1000$$

$\div 4$

$$4^n(n+1) > \frac{1000}{4}$$

$$\rightarrow \frac{500}{2} \rightarrow 250$$

$$4^n(n+1) > 250$$

$$\Rightarrow n > 7/3$$

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Try

$$n=1$$

$$4(2) = 8 < 250$$

$$n=2$$

$$4^2(3) = 16(3) = 48$$

$$n=3$$

$$4^3(4) = 64(4) = 256 > 250$$



$$\begin{array}{r} 3 \\ 16 \\ 16 \\ 1 \ 9 \ 6 \\ 16 \ 0 \\ \hline 2 \ 5 \ 6 \end{array}$$

7) Determine whether the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{9^n} \left(1 + \frac{2}{n}\right)^{n^2} \rightarrow a_n$$

Is absolute convergent, conditionally convergent or divergent.

$$a_n = \frac{(-1)^{n+1}}{9^n} \left(1 + \frac{2}{n}\right)^{n^2} \rightarrow |a_n| = \left| \frac{(-1)^{n+1}}{9^n} \left(1 + \frac{2}{n}\right)^{n^2} \right|$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{9^n} \left(1 + \frac{2}{n}\right)^{n^2}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \left[\frac{1}{9^n} \left(1 + \frac{2}{n}\right)^{n^2} \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{9} \left(1 + \frac{2}{n}\right)^n = \frac{1}{9} e^2 < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} |a_n|$ is convergent (Root test)

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is absolute convergent